FORM HS-SA · REV. 4 POST-STASIS REFRESHER · SUPPLEMENT A POST AT EVERY PERMIT BOARD & ENTRY HATCH

DUNEWRIGHT COLONY WORKS
Settlement Site 4 · Construction Division · Health & Safety

THE AIR CAN KILL YOU.

CONFINED SPACES LOOK HARMLESS. THE DANGER IS THE AIR INSIDE, AND YOU CAN'T SEE IT, SMELL IT OR FEEL IT.

WHAT COUNTS AS A CONFINED SPACE

ALL THREE = CONFINED SPACE

1 BIG ENOUGH TO GET INTO

Your whole body can enter and do work.

2 HARD TO GET OUT OF

Limited ways in or out: a hatch, a manway, a ladder, a crawl.

3 NOT BUILT FOR PEOPLE TO STAY IN

Nobody is meant to work there all shift.

It becomes a **permit-required space** if it has, or could have, a dangerous atmosphere, something that could engulf you (sand, grain, liquid, polymer), a shape that could trap or suffocate you, or any other serious hazard. **On this site, treat every confined space as permit-required until the entry supervisor says otherwise.** SITE RULE

TANKS & CISTERNS

Water, fuel and coolant tanks. Cisterns can hold low oxygen from rust or rot.

HULL VOIDS

Spaces between hab walls and decks. Often purged with inert gas.

DUCTS & PIPELINES

Air, water and service runs big enough to crawl through.

POLYMER VATS

Curing vapors, heat from the cure, and material that can trap you.

SEALED HABS

Unfinished modules with no air system running yet.

SAND HOPPERS

Flowing sand engulfs a person in seconds.

VAULTS & PITS

Utility vaults, sumps and pits. Heavy gases settle here.

TRENCH BOXES

Deep trenches and shafts with poor airflow.

TEST THE AIR FIRST

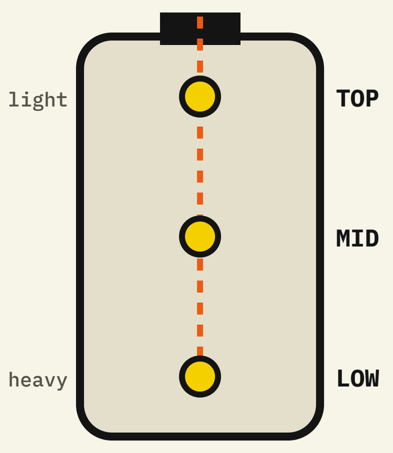
BEFORE ENTRY · THEN CONTINUOUSLY

1 · O_2 OXYGEN
19.5–23.5%
Under 19.5: too little to stay alert. Over 23.5: fire risk explodes.

2 · LEL FLAMMABLE
< 10%
Of the lower explosive limit. At 10% or above, no entry.

3 · H_2S
< 10 ppm
NIOSH ceiling. Old Earth OSHA ceiling 20 ppm. Kills your sense of smell.

3 · CO
< 35 ppm
NIOSH limit. Old Earth OSHA limit 50 ppm. No smell, no color.



1 **Oxygen first.** Most meters need enough oxygen to read flammables correctly.

2 **Flammables next.** Gas, fuel vapor, solvent and polymer off-gas.

3 **Toxics last.** H_2S, CO, and whatever the job brings, like welding fumes or resin vapor.

4 **Top, middle and bottom.** Light gases rise. Heavy gases like argon, H_2S and fuel vapor sink. Probe ahead of you as you go deeper.

5 **Keep monitoring.** Every entrant wears a running meter. It alarms: everyone gets out. No arguing.

INERT GAS

Argon and nitrogen are used to purge tanks, hull voids and weld zones. They push oxygen out.

ONE BREATH CAN DROP YOU.

In air with almost no oxygen, people collapse within seconds, often with no warning at all. You don't feel short of breath, because your body senses CO_2, not missing oxygen.

- A **purge tag** on a hatch means inert gas has been used inside. Never enter without a fresh test.
- **Ventilate** with forced fresh air until oxygen reads 19.5% or more at all levels, and keep the fan running.
- Argon is **heavier than air**. It pools in low spots, pits and the bottom of tanks.
- Never put your head into a hatch "just to look". That's the breath that kills.

FOUR ROLES, EVERY ENTRY

NAMED ON THE PERMIT BEFORE ANYONE GOES IN

ENTRANT

Goes inside. Wears a harness and meter. Knows the hazards and the signs of trouble. Gets out the moment the attendant says so, the meter alarms, or they feel wrong.

ATTENDANT

Stays outside the whole time. Keeps count of who's in. Stays in contact by voice or radio. **Never enters.** Calls rescue and starts the retrieval winch.

ENTRY SUPERVISOR

Checks the tests, the lockouts and the rescue plan. Signs the permit. Ends the permit when the job is done or conditions change.

RESCUE TEAM

Fire & Rescue (red helmets), on standby and trained for this space. Called on Ch 1 with "Confined space".

60%

About 6 in 10 people who die in confined spaces are would-be rescuers (NIOSH). They saw a coworker collapse and went in after them. The same air dropped them too. **Don't go in. Pull them out with the line, and call rescue.**

RESCUE WITHOUT GOING IN

1. Every entrant wears a **full-body harness** clipped to a **retrieval line**.
2. For vertical entries over 5 feet deep, the line runs to a **tripod or davit with a mechanical winch**.
3. Entrant down? The attendant calls **Ch 1: "Emergency, emergency, emergency. Confined space"** with location and how many are inside.
4. The attendant cranks them out with the winch from outside.
5. Only trained rescuers in supplied air go in if the line can't get them out.

BEFORE ANYONE ENTERS

- **Permit posted** at the hatch, with names, tests, times and the rescue plan.
- **Lock out** every pipe, pump, mixer, conveyor and power feed. See Module 7.
- **Ventilate** with forced fresh air. Keep exhaust and engines away from the intake.
- **Hot work inside?** It needs its own hot-work permit and fire watch. Welding eats oxygen and makes fumes. See Module 5.
- **Heat builds fast** in a tank in the sun. Use the hot-work heat level and timed entries. See Module 3.
- **Lighting and comms** tested. Use intrinsically safe lights where flammables are possible.

DO

- Read the permit before you go in.
- Watch your meter and your partner.
- Get out the first time something feels off: dizzy, headache, confused, out of breath.
- Tell the attendant every time you change position.

DON'T

- Enter without a test, a permit and an attendant.
- Use pure oxygen to "freshen" the air. It makes fires explosive.
- Unclip your retrieval line to reach further.
- Go in after someone who collapsed.

WORKER ACKNOWLEDGEMENT

I will not enter a confined space without a permit, a fresh air test, a harness on a retrieval line and an attendant outside. I will never go in after someone who collapsed.

NAME ____________________

COLONIST ID ____________________

CREW / SECTOR ____________________

DATE ____________________

Adapted from Old Earth standards: OSHA 29 CFR 1926 Subpart AA and 1910.146 permit-required confined spaces, NIOSH confined space guidance and recommended exposure limits, OSHA permissible exposure limits.

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